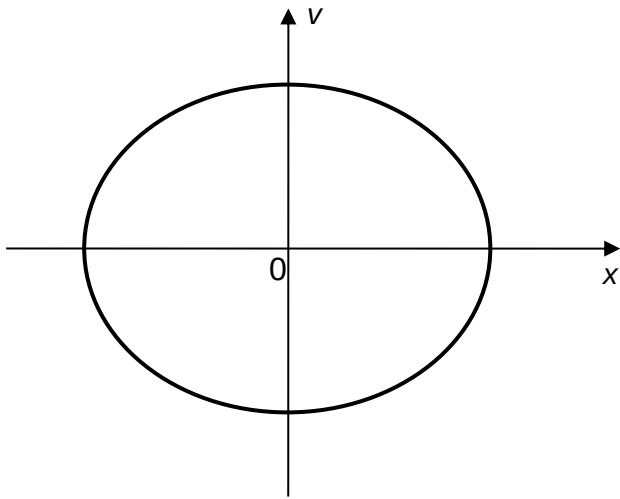
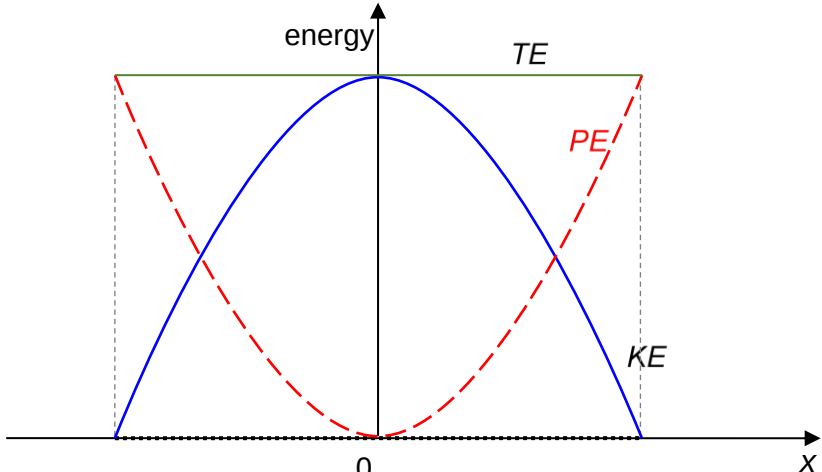


Qns	Answer	Marks
6(a)	acceleration proportional to displacement (from a fixed point)	B1
	either acceleration and displacement in opposite directions or acceleration always directed towards a fixed point	B1
6(b)(i)	g and r are constant so a is proportional to x	B1
	negative sign shows a and x are in opposite directions	B1
6(b)(ii)	$\omega^2 = \frac{g}{r}$ and $\omega = \frac{2\pi}{T}$	M1
	$\left(\frac{2\pi}{T}\right)^2 = \frac{9.81}{0.28}$	M1
	$T = 1.06 \text{ s}$	A0
6(b)(iii)	$x = x_o \sin \omega t$	
	$1.5 = 2.0 \sin\left(\frac{2\pi}{1.1} t\right)$	M1
	$t = 0.15 \text{ s}$ (or 0.143 s)	A1
6(b)(iv)	Correct shape	B1
		
	Correct maximum velocities Label units of v $v_o = \omega x_o$ $v_o = \left(\frac{2\pi}{T}\right) x_o$ $v_o = \left(\frac{2\pi}{1.1}\right)(0.020) = 0.11 \text{ m s}^{-1}$ (or 0.118 m s^{-1})	B1 B1
6(b)(v)	Total energy = max KE	B1
	$= \frac{1}{2}(0.100)(0.11)^2$	C1
	$= 0.61 \text{ mJ}$ (or 0.69 mJ)	A1

6(b)(vi)	Correct shape of TE, KE and PE  Label TE value and unit on axis	B1 B1
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6(b)(vii)	<u>METHOD 1:</u> $N_{28}T_{28} = N_{29}T_{29}$ $N_{28}\sqrt{r_{28}} = N_{29}\sqrt{r_{28}}$ AND $(N+1)\sqrt{28} = N\sqrt{29}$ $N = 56.5$ $t = (56.5)(1.1) \quad \text{or} \quad t = (56.5)(1.06)$ $= 62 \text{ s} \quad \quad \quad = 60 \text{ s}$ <u>METHOD 2:</u> $N_{28}T_{28} = N_{29}T_{29}$ $N\sqrt{28} = (N-1)\sqrt{29}$ $N = 57.5$ $t = (57.5)(2\pi\sqrt{\frac{0.29}{9.81}})$ $= 62 \text{ s}$ <u>METHOD 3:</u> $\Delta\phi = 2\pi$ $\phi - \phi' = 2\pi$ $\omega t - \omega' t = 2\pi$ $\left(\frac{2\pi}{T}\right)t - \left(\frac{2\pi}{T'}\right)t = 2\pi$ $\left(\frac{2\pi}{1.1}\right)t - \left(\frac{2\pi}{1.08}\right)t = 2\pi \quad \text{OR} \quad \left(\frac{2\pi}{1.06}\right)t - \left(\frac{2\pi}{1.08}\right)t = 2\pi$ $t = 59 \text{ s} \quad \quad \quad t = 57 \text{ s}$	C1 C1 C1 A1 C1 C1 A1 C1 C1 C1 A1
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	<u>METHOD 4:</u> $\Delta\phi = 2\pi$ $\phi - \phi' = 2\pi$ $\omega t - \omega' t = 2\pi$ $\left(\sqrt{\frac{g}{r}}\right)t - \left(\sqrt{\frac{g}{r'}}\right)t = 2\pi$ $\left(\sqrt{\frac{9.81}{0.28}}\right)t - \left(\sqrt{\frac{9.81}{0.29}}\right)t = 2\pi$ $t = 61 \text{ s}$ <u>METHOD 5:</u> $\Delta T = 1.08 - 1.06 = 0.02 \text{ s}$ $\frac{1.06}{0.02} = 53$ $t = 53 \times 1.06$ $t = 56 \text{ s}$	C1 C1 C1 A1 C1 C1 C1 A1
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Qns	Answer	Marks
7 (c)	work done per unit mass	D1